

## Solution Architecture

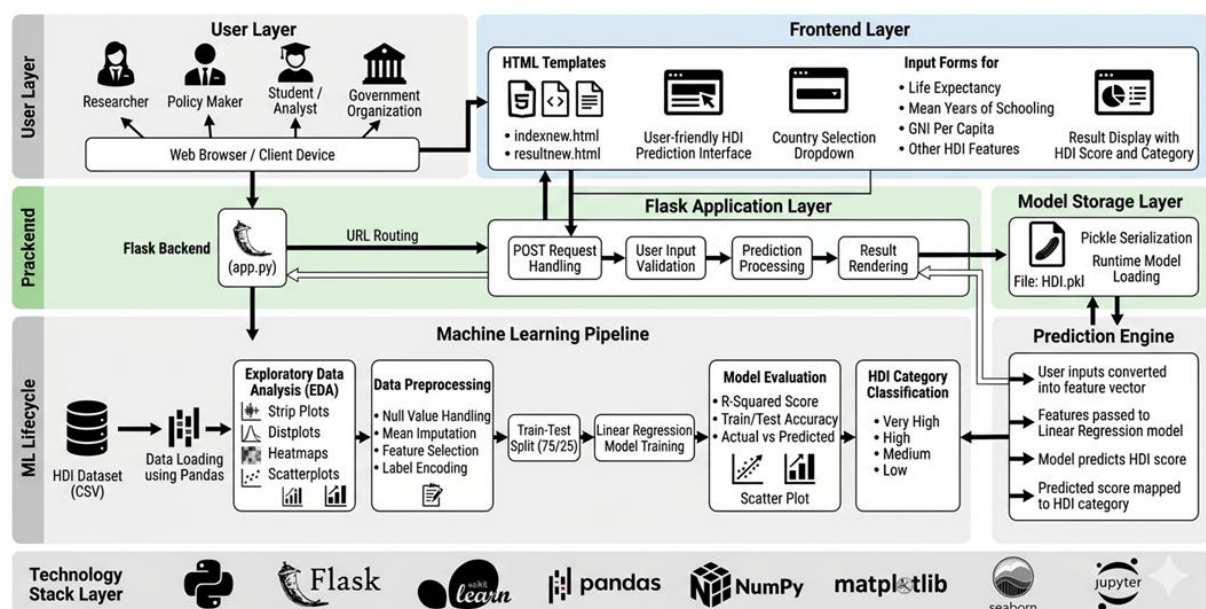
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|----------------------|---------------------------------------|
| <b>Date</b>          | 29 June 2026                          |
| <b>Team ID</b>       |                                       |
| <b>Project Name</b>  | A Comprehensive Measure of Well-Being |
| <b>Maximum Marks</b> | 2 Marks                               |

### Solution Architecture:

The **Solution Architecture** for the **Human Development Index (HDI) Predictor** project provides an intelligent framework for predicting a country's Human Development Index (HDI) score and development category using Machine Learning. Its goals are to:

1. Develop an accurate Human Development Index prediction system using historical HDI datasets.
2. Define the workflow between the dataset, Machine Learning model, and Flask web application.
3. Process user inputs such as Life Expectancy, Mean Years of Schooling, Expected Years of Schooling, and GNI per Capita to generate real-time HDI predictions.
4. Provide a scalable and user-friendly web application for governments, researchers, policymakers, and development organizations to analyze human development efficiently.

### Example - Solution Architecture Diagram:



## Component Description Table

| Component Name                | Description / Role in Architecture                                                                                                                                                                               | Technologies Used                                                   |
|-------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------|
| <b>Presentation Layer</b>     | Provides the user interface where users enter Life Expectancy, Mean Years of Schooling, Expected Years of Schooling, and GNI per Capita to predict the Human Development Index and view prediction results.      | HTML, CSS, JavaScript                                               |
| <b>API Gateway</b>            | Receives requests from the presentation layer, validates user inputs, preprocesses the data, loads the trained Machine Learning model, generates HDI predictions, and returns the results to the user interface. | Flask, Python, Pickle                                               |
| <b>Machine Learning Model</b> | Hosts the trained Linear Regression model that processes input features and predicts the HDI score and corresponding development category (Very High, High, Medium, or Low).                                     | Scikit-learn, Python, hdi_model.pkl                                 |
| <b>Data Processing Layer</b>  | Stores and processes the HDI dataset, performs data cleaning, feature engineering, visualization, train-test splitting, model training, evaluation, and saves the trained model.                                 | Pandas, NumPy, Matplotlib, Seaborn, Scikit-learn, HDI Dataset (CSV) |